

ANTON ALYAKIN

alyakin314@gmail.com $\diamond$ alyakin314.github.io

EDUCATION

Washington University in St. Louis

Doctor of Medicine

2024 Congress of Neurological Surgeons AI Fellow

[Aug 2021, May 2027]

(Expected)

Johns Hopkins University

Master of Science in Engineering

Applied Mathematics & Statistics

[Jan 2019, Dec 2019]

Johns Hopkins University

Bachelor of Science

Computer Science

Applied Mathematics & Statistics

[Aug 2015, May 2019]

RESEARCH

NYU Langone Health

AI Research Engineer

Neurosurgery / OLAB

[Mar 2024, Present]

Washington University in Saint Louis

Medical Student Researcher

Neurosurgery / Leuthardt Lab

[Mar 2023, Present]

Johns Hopkins University

Assistant Research Engineer

Applied Mathematics & Statistics / Neurodata Lab

[Jan 2020, Mar 2021]

Graduate Research Assistant

Applied Mathematics & Statistics / Priebe Group

[Jun 2019, Dec 2019]

Undergraduate Research Assistant

Computer Science / Saria Lab

[May 2017, May 2019]

SKILLS

Languages (in order of proficiency):

Python, English, Russian, R, Matlab, Java, C++.

Python skills:

PyTorch[DDP, FSDP, DeepSpeed, PyTorchLightning, LightningFabric, HuggingFaceAccelerate],
Tensorflow, vLLM, API[OpenAI, Anthropic, AWS Bedrock].

Other skills:

L^AT_EX, Git, Databases[PostgreSQL, BigQuery], HPC[IBM Spectrum LSF, SLURM], Docker,
OpenClaw, Fly.io, W&B, Lead climbing (6b/5.10), Bouldering (7a/V6 indoor; V2 outdoor).

DISSERTATIONS

1. **A. Alyakin**, *Robust Hypothesis Testing of Location Parameters using Lq-Likelihood-Ratio-Type Test in Python*, a thesis submitted to The Johns Hopkins University in conformity with the requirements for the degree of Master of Science in Engineering, 2019. [arXiv] [library] [code]
2. **A. Alyakin**, *Motif Discovery in the Irregularly Sampled Time Series Data*, a thesis submitted to The Johns Hopkins University in conformity with the requirements for Senior Honors Thesis in Computer Science, 2019. [osf] [pdf]

PUBLICATIONS

1. **A. Alyakin**, J. Stryker, D. A. Alber, J. V. Lee, K. L. Sangwon, B. Duderstadt, A. Save, D. Kurland, S. Frome, S. Singh, J. Zhang, E. Yang, K. Y. Park, C. Orillac, A. Valliani, S. Neifert, A. Liu, A. Patel, C. Livia, D. Lau, I. Laufer, P. A. Rozman, E. T. Hidalgo, H. Riina, R. Feng, T. Hollon, Y. Aphinyanaphongs, J. G. Golfinos, L. Snyder, E. Leuthardt, D. Kondziolka, E. K. Oermann, *CNS-Obsidian: A Neurosurgical Vision-Language Model Built From Scientific Publications*, Neurosurgery, to appear, 2026. [arXiv] [code] [model]
2. **A. Alyakin**, D. Alber, J. V. Lee, J. Stryker, S. Singh, A. Save, D. Kurland, C. Orillac, A. Valliani, S. Neifert, D. Lau, I. Laufer, P. A. Rozman, E. T. Hidalgo, H. Riina, L. Snyder, E. C. Leuthardt, D. Kondziolka, E. K. Oermann, *Automated Generation & Human Evaluation of Neurosurgical Board Examination Self-Assessment Questions*, Neurosurgery Practice, to appear, 2025.
3. **A. Alyakin**, J. Stryker, J.V. Lee, R. Feng, T. Hollon, D. Kondziolka, E. K. Oermann, *AI-Powered Pipeline Transforms Neurosurgical Articles into High-Quality Graphical Abstracts*, Neurosurgery Practice, to appear, 2025.
4. N. Maarouf, D. Reinecke, A. Smith, J. E. Markert, T. G Cogan, X. Han, **A. Alyakin**, D. A. Alber, M. Park, N. K. Goff, H. Weiss, E. S. Harake, K. Eddy, T. Hollon, E. K. Oermann, D. A. Orringer, *Natural Language Processing Methods Automate Molecular Marker Extraction from Glioma Pathology Reports*, Neurosurgery, to appear, 2025.
5. S. F. Chen, **A. Alyakin**, A. Seas, E. Yang, J. Choi, J. V. Lee, A. Chen, P. Warman, R. Bitolas, R. J. Steele, D. Alber, E. K. Oermann, *LLM-assisted Systematic Review of Large Language Models in Clinical Medicine*, Nature Medicine, 2026. [journal] [code]
6. K. Vishwanath, **A. Alyakin**, M. Ghosh, J. V. Lee, D. A. Alber, K. Sangwon, D. Kondziolka, E. K. Oermann, *Evaluating the Performance and Fragility of Large Language Models on Neurosurgical Board-like Questions*, Neurosurgery, 2025. [journal] [arXiv]
7. X. Han^{†1}, **A. Alyakin**[†], S. Ciprut, C. Lapierre, J. Stryker, J. Golfinos, D. Kondziolka, E. K. Oermann, *Neuro Data Hub: A New Approach for Streamlining Medical Clinical Research*, Neurosurgery Practice, 2025. [journal]
8. K. Vishwanath, J. Stryker, **A. Alyakin**, D. A. Alber, E. K. Oermann, *MedMobile: a Mobile-sized Language Model with Clinical Capabilities*, BMJ Digital Health & AI, 2025. [journal] [arXiv] [code]
9. S. Singh, **A. Alyakin**, D. A. Alber, J. Stryker, A. P. S Tong, K. Sangwon, N. Goff, M. de la Paz, M. Hernandez-Rovira, K. Y. Park, D. Kondziolka, E. K. Oermann, *The Pitfalls of Multiple-Choice Questions in Generative AI and Medical Education*, Scientific Reports, 2025. [journal] [arXiv]
10. K. L. Sangwon, X. Han, A. Becker, Y. Zhang, R. Ni, J. Zhang, D. A. Alber, **A. Alyakin**, M. Nakatsuka, N. Fabbri, Y. Aphinyanaphongs, J. Yang, A. Chachoua, D. Kondziolka, I. Laufer, E. K Oermann, *Automating the Referral of Bone Metastases Patients With and Without the Use of Large Language Models*, Neurosurgery, 2025. [journal]
11. **A. Alyakin**, D. Kurland, D. Alber, K. Sangwon, D. Li, A. Tsirigos, E. Leuthardt, D. Kondziolka, E. Oermann, *CNS-CLIP: Transforming a Neurosurgical Journal into a Multimodal Medical Model*, Neurosurgery, 2025. [journal]
12. D. A. Alber, Z. Yang, **A. Alyakin**, E. Yang, J. Zhang, S. Rai, A. A. Valliani, G. R. Rosenbaum, A. K. Amend-Thomas, D. B. Kurland, C. M. Kremer, A. Eremiev, B. Negash, D. D. Wiggan, M. A. Nakatsuka, K. L. Sangwon, S. Neifert, H. A. Khan, A. V. Save, A. Palla, E. A. Grin, M. Hedman, M. Nasir-Moin, C. X. Liu, L. Y. Jiang, M. A. Mankowski, D. L. Segev, Y. Aphinyanaphongs, H. A. Riina, J. G. Golfinos, D. A. Orringer, D. Kondziolka, E. K. Oermann, *Medical Large Language Models are Vulnerable to Data-poisoning Attacks*, Nature Medicine, 2025. [journal] [code]

^{1†} symbolizes equal contribution; order preserved as in manuscript.

13. R. Guennoun[†], **A. Alyakin**[†], H. Higushi, S. Demehri, *Commensal HPVs Have Evolved to Be More Immunogenic Compared with High-Risk α -HPVs*, Vaccines, 2024. [journal]
14. J. Lee, M. A. Ruiz-Cardozo, R. P. Patel, S. Javeed, R. S. Lavandi, C. Newsom-Stewart, **A. Alyakin**, C. A. Molina, N. Agarwal, W. Z. Ray, M. Santacatterina, B. H. Pennicooke, *Clinical Prediction for Surgical versus Nonsurgical Interventions in Patients with Vertebral Osteomyelitis and Discitis*, Journal of Spine Surgery, 2024. [journal]
15. **A. A. Alyakin**, J. Agterberg, H. S. Helm, and C. E. Priebe, *Correcting a Nonparametric Two-sample Graph Hypothesis Test for Graphs with Different Numbers of Vertices with Applications to Connectomics*, Applied Network Science, 2024. [journal] [arXiv] [code]
16. M. Powell, C. Clark, **A. Alyakin**, J. T. Vogelstein, B. Hart, *Exploration of Residual Confounding in Analyses of Associations of Metformin Use and Outcomes in Adults With Type 2 Diabetes*, JAMA Network Open, 2022. [journal] [arXiv]
17. F. Rahman, N. Finkelstein, **A. Alyakin**, N. A. Gilotra, J. Trost, S. P. Schulman, S. Saria, *Using Machine Learning for Early Prediction of Cardiogenic Shock in Patients with Acute Heart Failure*, Journal of the Society for Cardiovascular Angiography & Interventions, 2022. [journal] [arXiv]
18. K. Marchisio, Y. Park, A. Saad-Eldin, **A. Alyakin**, K. Duh, C. Priebe, P. Koehn, *An Analysis of Euclidean vs. Graph-Based Framing for Bilingual Lexicon Induction from Word Embedding Spaces*, Findings of the Association for Computational Linguistics: EMNLP 2021. [journal] [arXiv] [code]
19. J. Chung[†], B. Varjavand[†], J. Arroyo, **A. Alyakin**, J. Agterberg, M. Tang, J. T. Vogelstein, C. E. Priebe, *Valid Two-Sample Graph Testing via Optimal Transport Procrustes and Multiscale Graph Correlation with Applications in Connectomics*, Stat, 2021. [journal] [arXiv] [code]

PREPRINTS

1. A. Kapoor, **A. Alyakin**, J. E. Markert, A. Arias, M. Sughrue, E. K. Oermann, *Big Data in Neuro-Oncology*, 2026.
2. Y. Hong, S. Son, H. Jung, K. Vishwanath, **A. Alyakin**, J. V. Lee, E. K. Oermann, M. Sung, *Learning to Ignore: Enhancing Medical QA robustness under Unseen Distractors*, submitted, 2026.
3. J. V. Lee, **A. Alyakin**, A. L. Huguenard, R. G. Dacey, J. W. Osburn, *Preprocedural Quantitative Curvature Assessment Can Stratify Recanalization Risk Before Device Selection in Mechanical Thrombectomy*, 2026.
4. A. K. Kapoor, **A. Alyakin**, J. V. Lee, E. Yang, K. Vishwanath, A. M. Schulze, J. Lee, Y. Aphinyanaphongs, H. Riina, J. A. Frontera, E. K. Oermann, *Large Language Models for Predicting Functional Outcomes after Acute Ischemic Stroke*, submitted, 2026. [arXiv]
5. L. Y. Jiang[†], A. Chen[†], X. Han, X. C. Liu, R. Dua, K. Eaton, F. Wolff, R. Steele, J. Zhang, **A. Alyakin**, Q. Pan, Y. Chen, K. L. Sangwon, D. A. Alber, J. Stryker, J. V. Lee, Y. Aphinyanaphongs, K. Cho, E. K. Oermann, *Generalist Foundation Models Are Not Clinical Enough for Hospital Operations*, submitted, 2026. [arXiv]
6. K. Vishwanath, M. Ghosh, **A. Alyakin**, D. Alber, Y. Aphinyanaphongs, E. Oermann, *Generalist Large Language Models Outperform Clinical Tools on Medical Benchmarks*, 2025. [arXiv]
7. R. C. Santarosa, M. Anderson, C. Davison, R. N. Hagstrom, M. Scalfani, **A. Alyakin**, J. Lipman, C. Livia, A. Save, H. Weiss, E. Ponchione, V. Sharashidze, C. Y. Chung, J. Baranoski, C. W. Rutledge, E. Raz, E. Nossek, M. Shapiro, P. K. Nelson, H. A. Riina, *Surgical Clipping as a Successful Retreatment for WEB Neck Remnant: case report*, 2025.
8. A. K. Kapoor, **A. Alyakin**, R. Santarosa, C. William, S. Hoda, Y. Wang, A. Garrison, I. Littig, H. Weiss, E. K. Oermann, *Hemorrhagic Synovial Cyst Mimicking a Neoplastic Process: an illustrative case*, 2026.

9. K. Vishwanath, **A. Alyakin**, J. V. Lee, A. Anand, D. A. Alber, E. K. Oermann, *Medical Large Language Models are Easily Distracted*, 2025. [arXiv] [code]
10. **A. Alyakin**, “All You Need” Is Not All You Need for a Paper Title: On the Origins of a Scientific Meme, 2025. [arXiv] [code]
11. C. Hang, R. Deng, L. Y. Jiang, Z. Yang, D. A. Alber, **A. Alyakin**, E. K. Oermann, *BPQA Dataset: Evaluating How Well Language Models Leverage Blood Pressures to Answer Biomedical Questions*, 2025. [arXiv]
12. **A. Alyakin**, Y. Qin, and C. E. Priebe, *LqRT: Robust Hypothesis Testing of Location Parameters using Lq-Likelihood-Ratio-Type Test in Python*, 2019. [arXiv] [code]

PRESENTATIONS

1. **A. Alyakin**, J. Stryker, D. A. Alber, J. V. Lee, E. Leuthardt, D. Kondziolka, E. K. Oermann, *CNS-Obsidian: A Neurosurgery-Tailored Vision-Language Model Fine-Tuned Exclusively on Scientific Publications*, Oral Presentation at the 75th Annual Meeting of the Congress of the Neurological Surgeons, 2025. [poster]
2. X. Han[†], **A. Alyakin**[†], S. Ciprut, C. Lapierre, J. Stryker, A. Lowe, J. Golfinos, D. Kondziolka, E. K. Oermann, *Neuro Data Hub: A New Approach For Streamlining Medical Clinical Research*, Oral Presentation at the 75th Annual Meeting of the Congress of the Neurological Surgeons, 2025. [poster]
3. K. Vishwanath, **A. Alyakin**, M. Ghosh, J. V. Lee, D. A. Alber, K. Sangwon, D. Kondziolka, E. K. Oermann, *Evaluating the Performance and Fragility of Large Language Models on Neurosurgical Board-like Questions*, Mini Oral Presentation at the 75th Annual Meeting of the Congress of the Neurological Surgeons, 2025. [poster]
4. **A. Alyakin**, *Computational Optimization Of Laser Interstitial Thermal Therapy Parameters For Brain Tumor Ablation: A Thermodynamics-Based Simulation Approach*, Mini Oral Presentation at the 75th Annual Meeting of the Congress of the Neurological Surgeons, 2025. [poster]
5. **A. Alyakin**, J. Stryker, S. Singh, D. A. Alber, A. Save, D. Kurland, C. Orillac, A. A. Valliani, S. Neifert, D. Lau, I. Laufer, P. A. Rozman, E. T. Hidalgo, H. Riina, L. Snyder, E. C. Leuthardt, D. Kondziolka, E. Oermann, *Automated Generation And Human Evaluation Of Neurosurgical Board Examination And Self-Assessment Questions*, Mini Oral Presentation at the 75th Annual Meeting of the Congress of the Neurological Surgeons, 2025. [poster]
6. A. Y. Fu, A. A. Valliani, **A. Alyakin**, D. Kondziolka, E. K. Oermann, *Tabular Foundation Models Predict Clinical Outcomes in Skull Base and Cerebrovascular Surgery*, Mini Oral Presentation at the 75th Annual Meeting of the Congress of the Neurological Surgeons, 2025. [poster]
7. A. Save, **A. Alyakin**, J. Stryker, D. A. Alber, J. Zhang, G. Rosenbaum, S. Singh, E. Leuthardt, D. Kondziolka, E. K. Oermann, *A Platform for Serving Large Language Models for Neurosurgical Diagnostic Support*, 93rd American Association of Neurological Surgeons Annual Scientific Meeting, 2025.
8. K. L. Sangwon, **A. Alyakin**, D. Kurland, E. Leuthardt, D. Kondziolka, E. K. Oermann, *A Generalizable Pipeline for Building an Extensive Domain-Specific Dataset from a Medical Journal - Neurosurgery Edition*, Oral Presentation at the 74th Annual Meeting of the Congress of the Neurological Surgeons, 2024.
9. **A. Alyakin**, D. Kurland, D. Alber, K. Sangwon, D. Li, A. Tsirigos, E. Leuthardt, D. Kondziolka, E. Oermann, *CNS-CLIP: Transforming a Neurosurgical Journal into a Multimodal Medical Model*, Oral Presentation at the 74th Annual Meeting of the Congress of the Neurological Surgeons, 2024.

POSTERS

1. **A. Alyakin**, J. Stryker, J. V. Lee, R. Feng, T. Hollon, D. Kondziolka, E. K. Oermann, *AI-Powered Pipeline Transforms Neurosurgical Articles into High-Quality Graphical Abstracts*, Digital Poster at the 75th Annual Meeting of the Congress of the Neurological Surgeons, 2025. [poster]
2. **A. Alyakin**, J. V. Lee, H. Riina, E. K. Oermann, *Computational Optimization Of Endovascular Coil Deployment Parameters For Aneurysm Treatment: A Physics-Based Simulation Approach*, Digital Poster at the 75th Annual Meeting of the Congress of the Neurological Surgeons, 2025. [poster]
3. S. Singh, J. Zhang, **A. Alyakin**, E. K. Oermann, *GRPO Versus SFT: Optimizing Large Language Models For Neurosurgical Decision Support*, Physical Poster at the 75th Annual Meeting of the Congress of the Neurological Surgeons, 2025. [poster]
4. A. Kapoor, **A. Alyakin**, L. Blinder, A. Korpus, E. Yang, E. Yang, R. J. Steele, A. M. Schulze, J. A. Frontera, E. K. Oermann, *Large Language Models Predict Modified Rankin Scale After Acute Ischemic Stroke*, Digital Poster at the 75th Annual Meeting of the Congress of the Neurological Surgeons, 2025. [poster]
5. G. R. Rosenbaum, L. Y. Jiang, I. Sheth, J. Stryker, **A. Alyakin**, D. A. Alber, N. K. Goff, Y. J. F. Kwon, J. E. Markert, M. Nasir-Moin, J. M. Niehues, K. L. Sangwon, E. Yang, E. K. Oermann, *MedG-KRP: Medical Graph Knowledge Representation Probing*, Findings of the 4th Machine Learning for Health symposium (ML4H), 2024. [arXiv]
6. K. L. Sangwon, D. Kurland, **A. Alyakin**, D. Kondziolka, E. K. Oermann, *Seven Decades Of Change: Tracing The Evolution Of Neurosurgery Through Lexical Analysis Of Neurosurgery Publications Of The CNS (1955-2024)*, Digital Poster at the 74th Annual CNS Meeting, 2024.

TALKS

Congress of Neurological Surgeons

CNS 2025 Special Symposium 3: Data Science and AI for Neurosurgeons Oct 11th, 2025
Vision-Language Models

2025 Artificial Intelligence in Neurosurgery: A Practical Course May 31st, 2025
How Can We Build Foundation Models with Neurosurgical Data

Global AI Frontier Lab

Global AI Frontier Lab: Seminar Series Jun 16th, 2025
Data Quality is Multidimensional: Product-Market-Data Fit in AI and Neurosurgery

REVIEW

Neurosurgery

Reviewer [2024, present]

Quantitative Imaging in Medicine and Surgery

Reviewer [2024, present]

Scientific Reports

Reviewer [2025, present]

npj: Digital Medicine

Reviewer [2025, present]

BMC Neurology

Reviewer [2025, present]

IEEE Transactions on Signal and Information Processing over Networks

Reviewer [2026, present]

AWARDS

Johns Hopkins University

Applied Mathematics & Statistics Prize for Outstanding Master's Research	2020
Applied Mathematics & Statistics Achievement Award	2019
Undergraduate Departmental Honors with Thesis, Computer Science	2019
Undergraduate Departmental Honors, Applied Mathematics & Statistics	2019
Undergraduate General Honors	2019
Whitening School of Engineering Dean's List (8/8 Semesters)	2015-2019

TASIS The American School In Switzerland

2015 Shah Akbar Khan Award for Excellence in Mathematics	2015
2015 AP Scholar with Distinction	2015
Valedictorian	2015

TEACHING

Congress of Neurological Surgeons

Faculty

CNS 2025 Special Symposium 3: Data Science and AI for Neurosurgeons	Oct 2025
2025 Artificial Intelligence in Neurosurgery: A Practical Course	May 2025

Teaching Assistant

CNS 2024 Special Symposium 3: Data Science and AI for Neurosurgeons	Sep 2024
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Johns Hopkins University

Teaching Assistant

580.475 Biomedical Data Science	Fall 2019
553.430/630 Introduction to Statistics	Spring 2019
553.436/636 Data Mining	Fall 2018

MENTORING

Joanne Choi

<i>Visiting Medical Student Researcher at NYU OLAB</i>	[2025, Present]
1 abstract submitted	

Anjali Kapoor

<i>Post-baccalaureate Researcher at NYU OLAB</i>	[2025, Present]
1 poster (CNS 2025)	
1 manuscript submitted	

Krithik Vishwanath

<i>Visiting Undergraduate Researcher at NYU OLAB</i>	[2024, Present]
2 manuscript published (Neurosurgery; BMJ Digital Health & AI)	
2 manuscripts submitted	
Churchill scholar	

Shrutika Singh

<i>Undergraduate Researcher at NYU OLAB</i>	[2024, Present]
1 manuscript published (Scientific Reports)	
1 oral presentation (CNS 2025)	
1 poster (CNS 2025)	

Gabriel Rosenbaum

<i>High School Student Researcher at NYU OLAB</i>	[2024, 2024]
1 poster (ML4H 2024)	

SOCIETIES

Congress of Neurological Surgeons

Medical Student Member

[2024, present]

CNS Quarterly Member Spotlight

Summer 2025

Sergeant-At-Arms

CNS 2025

American Association of Neurological Surgeons

Medical Student Member

[2021, present]

GROUPS

Washington University School of Medicine

WUSM Global Health and Medicine

Treasurer

[2022, 2022]

WUSM Global Surgery Student Alliance Chapter

Treasurer

[2022, 2022]

WUSM ClimbRx (Climbing Club)

Co-lead

[2022, 2022]